ADRIATICA D.O.O.

| | |
|---|---|
| Name of Ship | **M/T SAMPLE** |
| IMO No | **XXXXXXXX** |
| | |
| Name of Plan | **BIOFOULING MANAGEMENT PLAN** |
| Plan No | **ADRI-BMP-2026-R0** |

ADRIATICA D.O.O.
Engineering & Technical Consultancy
Podgorica, Montenegro | info@adriaticadoo.me | www.adriaticadoo.me

FRONT MATTER

1. DOCUMENT CONTROL

| Field | Information |
|---|---|
| **Document Title** | Biofouling Management Plan |
| **Document Number** | ADRI-BMP-2026-R0 |
| **Revision** | 0 |
| **Prepared by** | Adriatica D.O.O. |
| **Approved by** | [Company Technical Manager Name] |
| **Effective Date** | 15 March 2026 |
| **Next Review Date** | 15 March 2027 |

2. DISTRIBUTION LIST

| Copy No. | Holder | Location |
|---|---|---|
| 1 | Master | Onboard |
| 2 | Company Technical Manager | Office |
| 3 | SMS Library | Onboard |
| 4 | Flag Administration | (if required) |
| 5 | Class Society | (if required) |

3. AMENDMENT RECORD

| Rev. | Date | Description | Approved by |
|---|---|---|---|
| 0 | 15.03.2026 | Initial issue | Tech. Manager |
| | | | |
| | | | |
| | | | |
| | | | |

TABLE OF CONTENTS

| | PAGE |
|---|---|
| 1. INTRODUCTION | 4 |
| 1.1. Scope | 4 |
| 1.2. General | 4 |
| 1.3. Definitions | 5 |
| 1.4. Plan Review and Update | 7 |
| 2. VESSEL PARTICULARS | 8 |
| 2.1. General Particulars | 8 |
| 2.2. Principal Dimensions | 8 |
| 3. ANTI-FOULING STRATEGY | 10 |
| 4. DESCRIPTION OF VESSEL'S OPERATING PROFILE | 11 |
| 5. DESCRIPTION OF VESSEL'S AREAS SUSCEPTIBLE TO BIOFOULING | 14 |
| 6. OPERATION & MAINTENANCE OF VESSEL'S ANTI-FOULING SYSTEM | 20 |
| 6.1. Timing of Operational and Maintenance Activities | 20 |
| 6.2. Maintenance Procedures During Dry-Docking | 21 |
| 6.3. In-Water Cleaning and Maintenance Procedures | 22 |
| 6.4. Operation of Onboard Treatment Processes | 28 |
| 7. SAFETY PROCEDURES FOR SHIP AND CREW | 30 |
| 8. DISPOSAL OF BIOLOGICAL WASTE | 33 |
| 9. RECORDING REQUIREMENTS | 34 |
| 10. CREW TRAINING AND FAMILIARIZATION | 36 |

APPENDICES

| | |
|---|---|
| APPENDIX I | DESCRIPTION OF VESSEL'S ANTI-FOULING SYSTEMS |
| APPENDIX II | BIOFOULING RECORD BOOK FORM |
| APPENDIX III | DECISION SUPPORT TOOL FOR IN-WATER CLEANING |
| APPENDIX IV | TYPES OF ANTI-FOULING COATINGS |
| APPENDIX V | INFORMATION ON CURRENTLY AVAILABLE IN-WATER CLEANING TECHNOLOGY |
| APPENDIX VI | SPECIFIC CALIFORNIA STATE REQUIREMENTS |
| APPENDIX VII | SPECIFIC NEW ZEALAND REQUIREMENTS |
| APPENDIX VIII | SUPPORTING FORMS AND REPORTS |
| APPENDIX IX | QUICK GUIDE FOR CREW |
| APPENDIX X | RELEVANT INFORMATION |

# 1. INTRODUCTION

## 1.1 SCOPE

This Plan has been developed pursuant to the 2011 Guidelines for the Control and Management of Ships' Biofouling to Minimize the Transfer of Invasive Aquatic Species, Resolution MEPC.207(62), adopted on 15 July 2011, and the US Ballast Water Management Rule, 33 CFR 151.2050(g), adopted on 21 June 2012.

The latest specific requirements of California State and New Zealand have been also taken into account (refer to Appendices VI and VII respectively) when developing this Plan.

The aim of this Plan is to provide effective procedures and practical guidance to the vessel's master, operator and owner and any other interested parties, on biofouling management measures in order to minimize the risk of transferring invasive aquatic species from ships' biofouling. It is important that these biofouling management procedures be effective as well as environmentally safe, practical and designed to minimize costs and delays to the vessel.

The management measures outlined herein is intended to complement current maintenance practices carried out within the Company.

This Plan is to be available for viewing on request by a Port State Authority and is written in the English language which is the working language of the crew.

**Regulatory Cross-Reference Table**

| Regulation / Requirement | Applicable Section(s) |
|---|---|
| **IMO Biofouling Guidelines (MEPC.207(62))** | Section 3, Section 5, Section 6, Appendix II |
| **IMO AFS Convention (2001)** | Section 3, Appendix I |
| **US Ballast Water Management Rule (33 CFR 151.2050(g))** | Section 1.1 (Scope) |
| **California Code of Regulations (Title 2, Div. 3, Ch. 1, Art. 4.8)** | Appendix VI |
| **New Zealand Craft Risk Management Standard (CRMS)** | Appendix VII |
| **ABS Guide for Biofouling Management (2020)** | Section 4.2, Section 5, Section 6.1 |
| **DNV-RP-B101 (2019)** | Section 4.2, Section 6.3 |

**Note:** This Plan has been developed pursuant to the IMO Biofouling Guidelines (MEPC.207(62) as amended by MEPC.378(80)).

## 1.2 GENERAL

Studies have shown that biofouling can be a significant factor for the transfer of invasive aquatic species. Biofouling on ships may result in the establishment of invasive aquatic species which may pose threats to human, animal and plant life, economic and cultural activities and the aquatic environment.

While the International Convention on the Control of Harmful Anti-Fouling Systems on Ships, 2001 (AFS Convention) addresses anti-fouling systems on ships, its focus is on the prevention of adverse impacts from the use of anti-fouling systems and the biocides they may contain, rather than preventing the transfer of invasive aquatic species.

The potential for invasive aquatic species transferred through biofouling to cause harm has been recognized by the IMO, the Convention on Biological Diversity (CBD), several UNEP Regional Seas Conventions (e.g., Barcelona Convention for the Protection of the Mediterranean Sea Against Pollution), the Asia Pacific Economic Cooperation forum (APEC), and the Secretariat of the Pacific Region Environmental Program (SPREP).

All ships have some degree of biofouling, even those which may have been recently cleaned or had a new application of an anti-fouling coating system. Studies have shown that the biofouling process begins within the first few hours of a ship's immersion in water.

The biofouling that may be found on a ship is influenced by a range of factors, such as:

1. design and construction, particularly the number, location & design of niche areas;
2. specific operating profile, including factors such as operating speeds, ratio of time underway compared with time alongside, moored or at anchor, and where the ship is located when not in use (e.g., open anchorage or estuarine port);
3. places visited and trading routes; and
4. maintenance history, including: the type, age and condition of any anti-fouling coating system, installation and operation of anti-fouling systems and dry-docking / slipping and hull cleaning practices.

Implementing practices to control and manage biofouling can greatly assist in reducing the risk of the transfer of invasive aquatic species. Such management practices can also improve a ship's hydrodynamic performance and can be effective tools in enhancing energy efficiency and reducing air emissions from ships.

The procedures contained herein are intended to provide useful recommendations to the ship's crew and Company personnel on general measures to minimize the risks associated with biofouling for this particular ship.

To minimize the transfer of invasive aquatic species, the ship should implement biofouling management practices, including the use of anti-fouling systems and other operational management practices to reduce the development of biofouling.

The intent of such practices is to keep the ship's submerged surfaces, and internal seawater cooling systems, as free of biofouling as practical. Following this Plan and minimizing macrofouling would have a reduced potential for transferring invasive aquatic species via biofouling.

## 1.3. DEFINITIONS

**Adequate documentation** means records of the recent history of anti-fouling installation and hull maintenance undertaken on a ship or movable structure.

**AFS Convention** means the International Convention on the Control of Harmful Anti-Fouling Systems on Ships, 2001.

**Anti-fouling coating** means a coating used on a vessel to prevent or reduce the accumulation of biofouling. Common types of anti-fouling coating are described in Appendix IV.

**Anti-fouling coating system** means the combination of all component coatings, surface treatments (including primer, sealer, binder, anti-corrosive and anti-fouling coatings) or other surface treatments, used on a vessel to control or prevent attachment of unwanted aquatic organisms.

**Anti-fouling system** means a coating, paint, surface treatment, surface, or device that is used on a vessel to control or prevent attachment or association of unwanted organisms and / or biofouling.

**Anti-fouling System (AFS) Certificate** means the International Anti-fouling System Certificate that ships greater than 400 gross tons and registered to a Flag State that is a Party to the AFS Convention, 2001 are required to carry. This certificate indicates that the ship's anti-fouling system complies with the Convention.

**Biocide** means a chemical substance that is incorporated into anti-fouling coatings to prevent the settlement or survival of aquatic organisms. Biofouling, also referred to as hull fouling or marine growth, means the accumulation of aquatic organisms such as micro-organisms, plants, and animals on surfaces and structures immersed in or exposed to the aquatic environment, including, but not limited to, sea chests, propellers, anchors and associated chains, and other niche areas. Biofouling can include microfouling and macrofouling (see below).

**Biosecurity** means the exclusion, eradication or effective management of pests and diseases that threaten the economy, environment, human health, social and cultural values.

**Biosecurity risk** means the potential harm to the economy, environment, human health and social and cultural values

**Biofouling Risk Level** means the level of risk associated with biofouling accumulation based on the vessel's operating profile, trading areas, and coating condition, as defined in Section 4.2.

**Contaminant** means any undesirable substance occurring in the environment as a result of human activities, without adverse effects being observed.

**Contamination** means the presence of a contaminant in the environment, or the process whereby a contaminant is introduced into the environment.

**Controlled waste** means material or liquid waste that is regulated because of its toxicity or imminent hazardous nature.

**Declaration on Anti-fouling System** means the declaration required to be carried by vessels of less than 400 gross tons but greater than 24m, and registered to a Flag State that is Party to the AFS Convention, 2001. This declaration ensures that their anti-fouling coating system complies with the Convention.

**Exclusive Economic Zone (EEZ)** means those waters beyond the limits of the Territorial Sea out to 200 nautical miles.

**Fouling Severity / Fouling Rating** means the classification of biofouling extent on a scale from Level 0 (clean) to Level 3 (heavy macrofouling) as defined in Section 6.3.1.

**Hull Grooming** means regular, frequent cleaning of the hull to remove microfouling and prevent the establishment of macrofouling, typically using soft tools.

**In-water cleaning** means the physical removal of biofouling and / or anti-fouling coating surface deposits from a ship while in the water.

**In-water inspection** means underwater survey or inspection by divers (including inspections conducted with remotely operated vehicles). Inspections for purposes other
than surveying biofouling may be considered opportunities for evaluating biofouling extent.

**In-water treatment** means any method or process that is aimed at sterilizing biofouling from the wetted portions of a vessel while the vessel remains in the water.

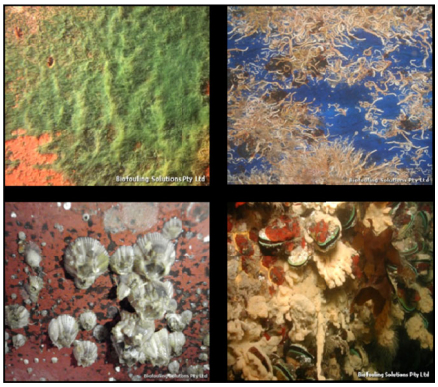
**Invasive aquatic species** means a species which may pose threats to human, animal and plant life, economic and cultural activities and the aquatic environment.

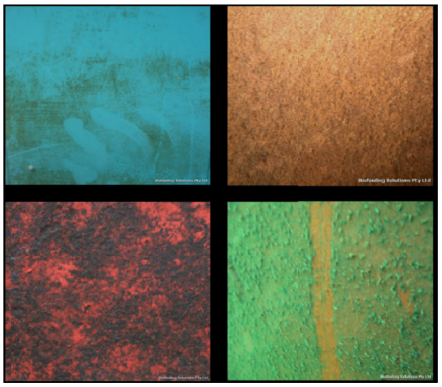
**Local water quality standard** means the concentrations or discharge of contaminants (such as those arising from hull maintenance operations) that are regarded as acceptable by the relevant authority.

**Maintenance facility** refers to any location or facility where on-shore maintenance of ships is carried out. This includes the maintenance, removal and application of anti-fouling coatings and the removal of biofouling organisms.

**Marine Growth Prevention System (MGPS)** means an anti-fouling system used for the prevention of biofouling accumulation in internal seawater cooling systems and sea chests and can include the use of anodes, injection systems and electrolysis.

**Member States** means States that are Members of the International Maritime Organization. Macrofouling means large, distinct multicellular organisms visible to the human eye such as barnacles, tubeworms, or fronds of algae and other large attached or mobile organisms.





**Microfouling** means microscopic organisms including bacteria and diatoms and the slimy substances that they produce. Biofouling comprised of only microfouling is commonly referred to as a slime layer and can be usually removed by gently passing a finger over the surface.

**Niche areas** mean areas on a vessel that may be more susceptible to biofouling due to different hydrodynamic forces, susceptibility to coating system wear or damage, or being inadequately, or not, painted, e.g., sea chests, bow thrusters, propeller shafts, inlet gratings, dry-dock support strips, etc.

**Niche Management** means the specific practices and procedures applied to manage biofouling in niche areas such as sea chests, thrusters, rudders, and other areas identified in Section 5.

**Organization** means the International Maritime Organization.

**Out-of-water maintenance** means removal of the vessel from the water and into a dry-dock or slipway for inspection or maintenance.

**Out-of-water support strips** means sections of the hull that rested on support blocks while the vessel was out of water in a dry-dock or slipway. These areas are typically not cleaned or treated with fresh anti-fouling systems, resulting in reduced antifouling protection.

**Percentage** cover means the percentage of the total surface area under examination that is occupied by biofouling.

**Planned in-service period** means the intended interval (decided at the time of antifouling coating application) until the next scheduled application of anti-fouling coating on a vessel.

**Port State authority** means any official or organization authorized by the Government of a port State to verify the compliance and enforcement of standards and regulations relevant to the implementation of national and international shipping control measures.

**Preventive Cleaning** means cleaning undertaken to prevent the accumulation of biofouling, typically at Level 1 or early Level 2, to avoid more extensive fouling. **Reactive Cleaning** means cleaning undertaken in response to observed fouling that has reached Level 2 or higher, or in response to operational or regulatory requirements.

**Relevant authority** means the authority that has responsibility for managing the environmental effects of activities.

**Service life** means the period of time an anti-fouling coating system is expected to protect a treated surface from biofouling and/or corrosion if the coatings are applied in accordance with the manufacturer's specifications.

**Ship means** a vessel of any type whatsoever operating in the aquatic environment and includes hydrofoil boats, air-cushion vehicles, submersibles, floating craft, fixed or floating platforms, floating storage units (FSUs) and floating production storage and off-loading units (FPSOs).

**Statement of Compliance** means a document (and associated evidence) issued by a Classification Society to ships greater than 400 gross tons that are registered in Flag States not Party to the AFS Convention, 2001.

**States** means coastal, port or Member States as appropriate.

**Treatment** means a process which may use a mechanical, physical, chemical or biological method to remove or render sterile, invasive or potentially invasive aquatic species fouling a ship.

**Waterline** means the area along the external hull of a vessel where the surface of the water interfaces with the air. The waterline is not a fixed location; its placement is dependent on loading and ballasting operations.

**Wetted portion of a vessel** means all parts of a vessel's hull and structures that are either submerged in water when the vessel is loaded to the deepest permissible legal draft or associated with internal piping structures in contact with water taken onboard.

## 1.4 PLAN REVIEW AND UPDATE

This Biofouling Management Plan shall be reviewed and updated:

- **Annually** by the Company technical department
- After each **dry-docking** (to reflect coating changes, hull modifications)
- Following any **significant biofouling event** or non-conformity
- When **regulatory requirements** change (IMO, port State, flag State)
- When the vessel's **operating profile** changes significantly

All reviews shall be documented in the Amendment Record. Updated versions shall be distributed to all holders and obsolete copies removed.

# 2. VESSEL PARTICULARS

## 2.1 GENERAL PARTICULARS

| FIELD / SPECIFICATION | DETAILS & DATA |
|---|---|
| Ship's Name | |
| Ship's Type | |
| Flag State | |
| Port of Registry | |
| Classification Society | |
| Call Sign | |
| MMSI | |
| IMO Number | |
| Gross Tonnage | |
| Net Tonnage | |
| Built by | |
| Year Built | |

## 2.2 PRINCIPAL DIMENSIONS

| FIELD / SPECIFICATION | DETAILS & DATA |
|---|---|
| Length O.A. | |
| Length B.P | |
| Breadth (mld.) | |
| Depth to Main Deck (mld.) | |

**Note:** Fill in the vessel's actual particulars. Example values are provided for guidance.